

the shared form: multiplication by φ on $\mathbb{Q}(\sqrt{5})$

naive T

$$\begin{bmatrix} 1/2 & 5/2 \\ 1/2 & 1/2 \end{bmatrix}$$

trace form $B = \text{diag}(2, 10)$

$$B T = \begin{bmatrix} 1 & 5 \\ 5 & 5 \end{bmatrix}$$

$$T \neq T^T$$

$$BT = (BT)^T \Rightarrow B\text{-self-adjoint}$$

same operator: non-symmetric naively, self-adjoint in the trace form
self-adjointness / positivity is FORM-RELATIVE